

9 Must-Know Python-Pandas Operations for Working with Data

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Data Import

```
pd.read_csv('file.csv')
pd.read_excel('file.xlsx', sheet_name='Sheet1')
pd.read_sql(query, connection)
pd.read_json('file.json')
pd.read_parquet('file.parquet')
```

Data Selection

```
df['column'] # Single column
df.loc['row', 'col'] # Label based
df.iloc[0:5, 0:2] # Integer based
df.query('col > 5') # SQL-like filtering
df[df['col'].isin(['A', 'B'])] # Multiple values
```

Data Manipulation

```
df.groupby('col').agg({'col2': ['mean', 'sum']})
df.merge(df2, on='key', how='left')
df.pivot_table(values='val', index='idx')
df.sort_values(['col1', 'col2'])
df.melt(id_vars='id', value_vars=['A', 'B'])
df.apply(lambda x: x*2) # Apply function
```

Statistics

```
df.describe() # Summary statistics
df['col'].agg(['mean', 'median', 'std'])
df['col'].value_counts(normalize=True)
df.corr(method='pearson')
df.cov() # Covariance matrix
df.quantile([0.25, 0.5, 0.75])
```

Data Cleaning

```
df.dropna(subset=['col'], how='any')
df.fillna(method='ffill') # Forward fill
df.drop_duplicates(subset=['col'])
df['col'].replace({'old': 'new'})
df['col'].astype('category')
df.interpolate(method='linear')
```

Time Series

```
df.resample('M').mean() # Monthly average
df.rolling(window=7).mean()
df.shift(periods=1) # Shift values
pd.date_range('2024', periods=12, freq='M')
df.asfreq('D', method='ffill')
df['date'].dt.strftime('%Y-%m-%d')
```

String Operations

```
df['col'].str.contains('pattern')
df['col'].str.extract('(\d+)')
df['col'].str.split('_').str[0]
df['col'].str.lower()
df['col'].str.strip()
df['col'].str.replace(r'\s+', ' ')
```

Advanced Features

```
df.pipe(func) # Method chaining
pd.eval('df1 + df2') # Expression eval
df.memory_usage(deep=True)
df.select_dtypes(include=['number'])
df.nlargest(5, 'col') # Top N values
df.explode('col') # Expand list column
```

Data Export

```
df.to_csv('output.csv', index=False)
df.to_excel('output.xlsx', sheet_name='Sheet1')
df.to_parquet('output.parquet')
df.to_json('output.json', orient='records')
```

Tips & Best Practices

- Use `.copy()` when creating DataFrame views
- Chain operations with method chaining
- Set `dtype='category'` for categorical columns
- Use `inplace=True` carefully, prefer reassignment